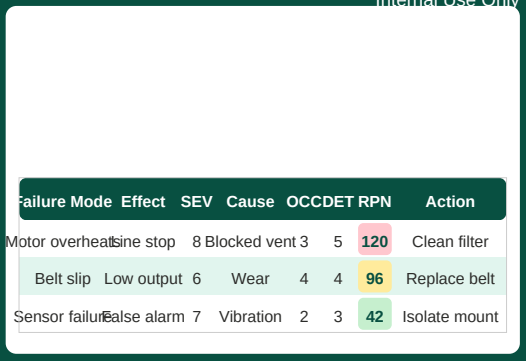
QUALITY & PROCESS

Internal Use Only

FMEA

Proactively identify potential failure modes, their effects, causes, and controls before they occur — then prioritise risk



Failure Mode	Effect	SEV	Cause	OCC	DET	RPN	Action
Motor overheats	Line stop	8	Blocked vent	3	5	120	Clean filter
Belt slip	Low output	6	Wear	4	4	96	Replace belt
Sensor failure	False alarm	7	Vibration	2	3	42	Isolate mount

WHEN TO USE

Before launching a new process or product; when designing controls for high-risk steps; after a serious incident.

HOW TO APPLY — STEPS

1. List all process steps or product functions
2. Identify potential failure modes for each step
3. Assess Severity, Occurrence, and Detection each on a 1–10 scale
4. Calculate RPN = Severity × Occurrence × Detection
5. Prioritise and action the highest RPN items first
6. Re-score after actions to confirm RPN has reduced

PRACTITIONER TIP

Prioritise HIGH Severity items regardless of RPN. Some failures are unacceptable no matter how rarely they occur.

KEY OUTPUTS

- FMEA table with RPN scores
- Risk priority ranking
- Action plan with verified revised RPNs

HOW TO MEASURE SUCCESS

Average RPN reduction %; number of critical failure modes with effective mitigation

TOOL CATEGORY

Quality & Process